

INTRODUCTION TO ROBOTICS FOR AI AND DATA SCIENCE

Topic 1: Introduction to Robotics and the AI Perspective



Dr. Akram Fatayer
Instructor



a.fatayer@zuj.edu.jo



Department of AI and Data Science
Al-Zaytoonah University of Jordan



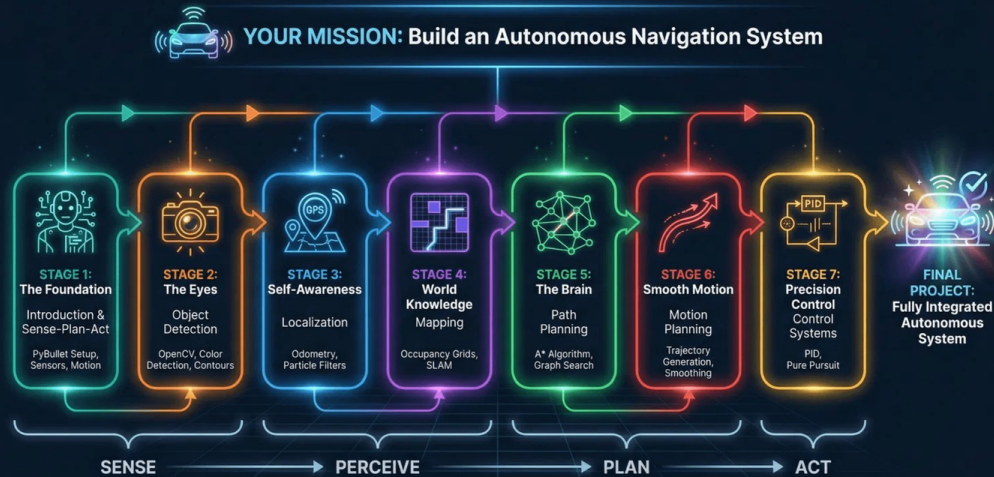
[linkedin.com/in/akram-fatayer](https://www.linkedin.com/in/akram-fatayer)

AL-ZAYTOONAH UNIVERSITY OF JORDAN

SENSE → PERCEIVE → PLAN → ACT

YOUR MISSION: BUILD AN AUTONOMOUS SYSTEM

From Vision to Reality: A Semester-Long Journey of Incremental Integration



WHAT IS A ROBOT?

A programmable machine capable of carrying out complex actions automatically by **adapting** to its environment.



SENSE

PERCEPTION

Gather data about the world and internal state.

Tools: Cameras, LiDAR, IMU, Encoders



DECIDE

DECISION MAKING

Process data to plan the best action.

Tools: Algorithms, AI, Path Planning



ACT

EXECUTION

Execute physical actions to change the world.

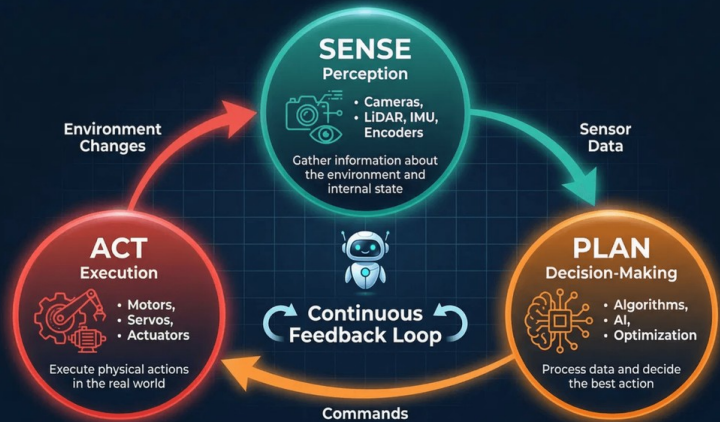
Tools: Motors, Servos, Hydraulics



Unlike simple automation, robots operate in a Continuous Feedback Loop

THE SENSE-PLAN-ACT CYCLE

The Foundation of All Robotic Systems



The Foundation of All Robotic Systems

SENSE

Gather data about the environment (Exteroceptive) and internal state (Proprioceptive).

PLAN

Process sensor data, update world model, and decide the optimal action.

ACT

Execute commands through motors and actuators to physically change the world.

The Feedback Loop: Actions change the environment, which changes the next sensor reading.

AUTONOMY EXISTS ON A SPECTRUM

From Human Control to Full Independence

6 Levels of Robot Autonomy



Course Focus: Levels 2-4



We build systems that operate autonomously in defined conditions (like a warehouse or a track) but may require human oversight for edge cases.

ROBOTICS IS RESHAPING INDUSTRIES

Robots Are Transforming Every Industry



Manufacturing

Industrial arms for welding, painting, assembly. Collaborative robots (cobots) working alongside humans.



Autonomous Vehicles

Waymo, Tesla, Cruise. Level 3-4 autonomy with sophisticated perception and planning.



Healthcare

da Vinci surgical robots, rehabilitation robots, assistive robots for elderly care.



Warehouse & Logistics

Amazon robots, Locus Robotics. Thousands of robots coordinating autonomously.



Agriculture

Autonomous tractors, harvesting robots, crop monitoring drones.



Space Exploration

Perseverance, Curiosity. Level 4-5 autonomy with minimal human intervention.

Robotics is not a niche field; it is the **physical interface of AI** transforming every major sector.

OUR DATA-FIRST APPROACH

Leveraging Your AI & Data Science Superpowers

Our Approach: Data-First Robotics

Traditional Approach



Start with unfamiliar concepts

Our Data-First Approach



Start with what you already know

We leverage your existing strengths in Python, Machine Learning, and Data Analysis.
You will see robotics through a Data Science lens.



We flip the script.
Instead of struggling with mechanics first, you apply your existing **Python** and **Machine Learning** skills to perception and planning immediately.

THE INCREMENTAL INTEGRATION APPROACH

How We Build Complex Intelligent Systems



1. VISION FIRST

Start with the end goal: The Autonomous Car.

Understand **WHY** before learning **HOW**.



2. DECOMPOSE

Break the complex system into manageable modules:

Perception, Localization,
Planning, Control.



3. BUILD BLOCKS

Master one module at a time through focused mini-projects.

(e.g., Build the Object
Detector).



4. INTEGRATE

Connect the new block to your growing system.

Verify interfaces and system
synergy.



We avoid the "Black Box" problem. You build every part, so you understand the whole.

ASSESSMENT STRUCTURE

Your Grade Reflects What You Build



PROJECT WORK

60%

- ✓ Weekly Labs & Mini-Projects
- ✓ Midterm Project Checkpoint
- ✓ Final Integrated System

FINAL EXAM

40%

- ✓ Conceptual Understanding
- ✓ Theoretical Foundations
- ✓ Algorithm Analysis

** Note: There is no traditional Midterm Exam. It is replaced by the Project Checkpoint.*

YOUR TOOLKIT

Industry-Standard Tools for Modern Robotics



Python 3.11

The Language of AI



PyBullet

Physics Simulation Engine



NumPy

Matrix Math & Linear Algebra



OpenCV

Computer Vision & Image Processing



Matplotlib

Data Visualization & Plotting



Simulation-First: Experiment freely, fail safely, and iterate quickly.

WHAT YOU WILL MASTER

Aligned with Program Learning Outcomes

KNOWLEDGE



MK

Simulation & Programming

Build and program robotic systems in simulation using Python and PyBullet, understanding core physics and software structure.

SKILLS



MS

Perception Pipelines

Process sensor data from cameras and encoders to detect objects and extract key features.



MS

Core Algorithms

Implement key robotics algorithms such as A* planning, SLAM, and PID control.

COMPETENCE



MC

1

System Integration

Integrate perception, planning, and control into a functioning autonomous navigation system.



MC2

Creative Solutions

Design innovative solutions to open-ended robotics challenges.



MT

1

Collaboration

Work effectively in teams to manage projects and communicate technical results.

TODAY'S LAB: MEET YOUR ROBOT

Stage 1: Setting the Foundation



The Goal:

Establish a working simulation environment and take control of your first autonomous agent.

1

Environment Setup

Configure Python 3.11 and install PyBullet, NumPy, and OpenCV.

2

Spawn Robot

Load a URDF model into the physics simulation world.

3

Read Sensors

Access real-time data: Position, Orientation, and Camera images.

4

Motion Control

Send velocity commands to drive the robot in a square pattern.

LET'S BUILD INTELLIGENT ROBOTS TOGETHER

YOUR MISSION: Build an Autonomous Navigation System

I am here to support your learning journey



INSTRUCTOR

Dr. Akram Fatayer



EMAIL

a.fatayer@zuj.edu.jo



OFFICE

Room 9354



LINKEDIN

linkedin.com/in/akram-fatayer

STAGE 1:
The Foundation

Introduction &
Sense-Plan-Act

PyBullet Setup,
Sensors, Motion

STAGE 3:
Self-Awareness

Localization

Odometry,
Particle Filters

STAGE 4:
World
Knowledge

Mapping

Occupancy Grids,
SLAM

STAGE 5:
Gain
Planning

Planning

A* Algorithm,
Graph Search

STAGE 6:
Smooth
Planning

Planning

Trajectory
Generation,
Smoothing

STAGE 7:
Precision
Control
Systems

Control
Systems

PID,
Pure Pursuit

FINAL
PROJECT:
Fully Integrated
Autonomous
System

"Whether you are struggling with a concept, excited about a project idea, or curious about research opportunities, **my door is open.**"

PLAN

ACT